over the vial. The hammer, in turn, would be connected to a Geiger counter, which detects radioactivity, and that counter would be pointed at a tiny lump of slightly radioactive metal. This Rube Goldberg contraption would be set off the moment the metal emitted any radiation; once that happens, the Geiger counter would register the radiation, which would release the hammer, smashing the vial and killing the cat. (Schrödinger had no intention of actually conducting this experiment, to the SPCA's relief.) Schrödinger proposed leaving the cat in the box for a certain period of time, then opening the box to find the cat's fate.

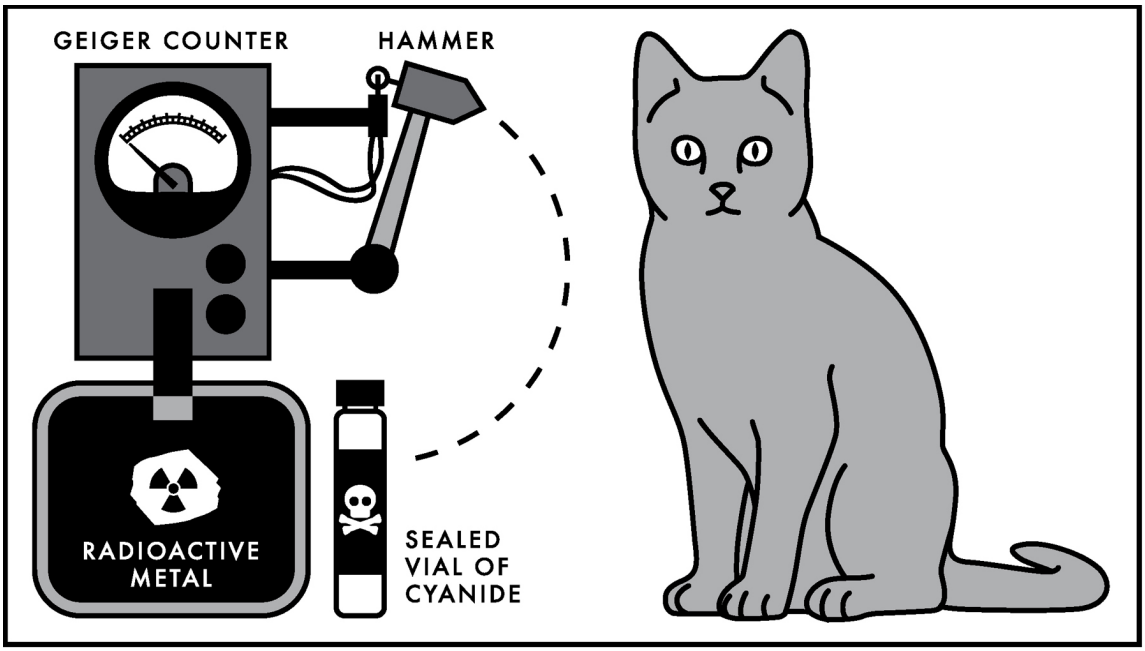


Figure I.1. Schrödinger's cat. When the metal gives off radiation, the Geiger counter will register it and drop the hammer, releasing the cyanide and killing the cat.

The radiation emitted by the lump of metal is composed of subatomic particles, breaking away from the atoms in the metal and flying off at high speeds. Like all sufficiently tiny things, those particles obey the laws of quantum physics. But, instead of reading Shakespeare, the subatomic particles in the metal have been listening to the Clash—at any particular moment, they don't know whether they should stay or they should go. So they do both: during the time the box is closed, the indecisive lump of